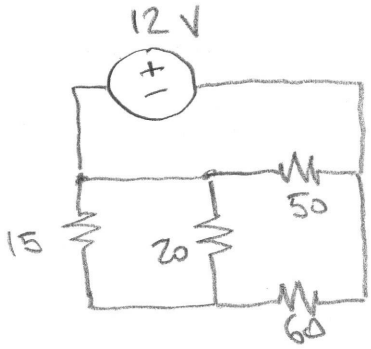
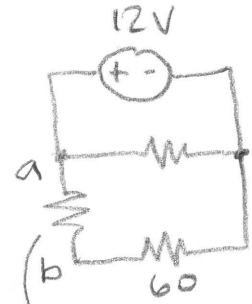


Fysisk PÅ Lösning
Till E0013E
2025-10-30

1.

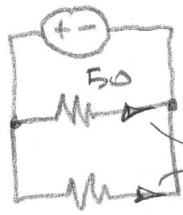


$\Rightarrow$



$$\frac{15 \cdot 20}{15+20} = 8.571\ \Omega$$

$\Rightarrow$



$$i_1 = \frac{12}{50} = 0.24\ A$$

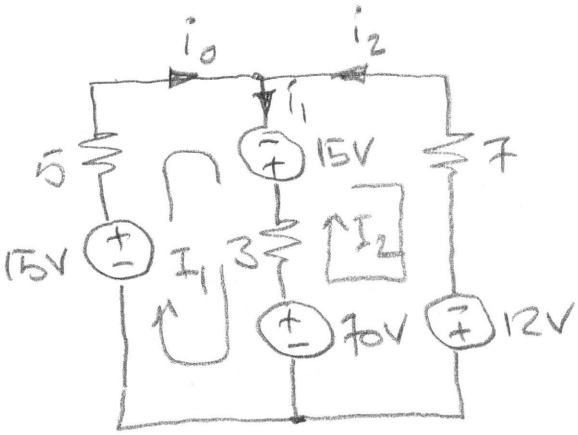
$$i_2 = \frac{12}{68.571} = 0.175\ A$$

SPÄNNINGSFALLET MELLAN a-b BLIR

$$V_{ab} = 8.571 - 0.175 = 1.4999 \simeq \underline{1.5\ V}$$

SVAR: SPÄNNINGEN BLIR $\underline{1.5\ V}$

2.



ANSÄTT FIKTIVA STRÖMMARNA I_1 OCH I_2. ANVÄND KSPL.

V: $15 - 5I_1 + 15 - 70 - 3(I_1 - I_2) = 0$

H: $7I_2 + 12 + 70 - 15 - 3(I_2 - I_1) = 0$

$$\Rightarrow \begin{cases} -40 - 8I_1 + 3I_2 = 0 \\ 67 - 10I_2 + 3I_1 = 0 \end{cases}$$

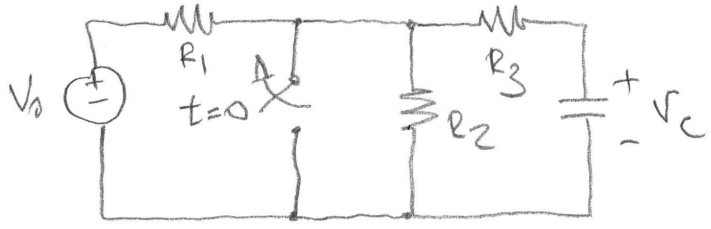
$$\Rightarrow \begin{bmatrix} -8 & 3 \\ 3 & -10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 40 \\ -67 \end{bmatrix}$$

SOM HAR LÖSNING $\begin{cases} I_1 = -\frac{199}{71} \\ I_2 = \frac{416}{71} \end{cases}$

VILKET GER

$$\begin{cases} i_0 = I_1 \approx -2.80\ \text{A} \\ i_1 = I_1 - I_2 \approx -8.66\ \text{A} \\ i_2 = -I_2 \approx -5.86\ \text{A} \end{cases}$$

3.



$V_0 = 24\,V$
$R_1 = 60\,k\Omega$
$R_2 = 30\,k\Omega$
$R_3 = 30\,k\Omega$
$C = 10\,nF$

a) för $t<0$ $\quad v_c(t<0) = 0$

för $t \to \infty$ gäller

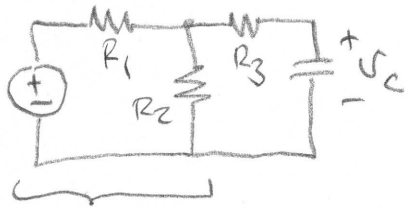


$$i_1 = \frac{V_0}{R_1+R_2}$$

$$V_{R_2} = R_2 i_1 = \frac{V_0 R_2}{R_1+R_2} = \underline{8\,V}$$

ALLTÅ $\quad V_{R_2} = v_c(t\to\infty) = \underline{8\,V}$

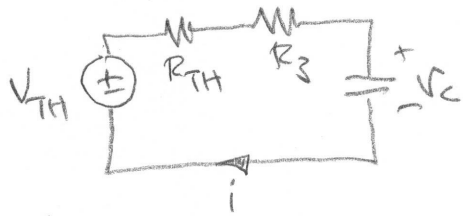
b) Bygg om LITE för $t \geq 0$



gör om till THEVENINEKV.

DÅR $V_{TH} = \frac{24\cdot 30}{60+30} = \underline{8\,V}$

$R_{TH} = \frac{60\cdot 30}{60+30} = \underline{20\,k}$



↑ ANVÄNDA KSPL

$$V_{TH} - (R_{TH}+R_3)i - v_c = 0 \qquad (1)$$

MEN $\quad v_c = \underbrace{v_c(0)}_{=0} + \frac{1}{C}\int i\,dt$

$$\Rightarrow V_{TH} - (R_{TH}+R_3)i - \frac{1}{C}\int i\,dt \quad \text{OBS}$$

DELIVERA MAPPT

$$(R_{TH}+R_3)\frac{di}{dt} + \frac{1}{C}i = 0$$

$$\frac{di}{dt} + \frac{1}{(R_{TH}+R_3)C}i = 0$$

SOM HAR LÖSNING

$$i(t) = Ae^{-t/\tau}, \quad \tau = (R_{TH}+R_3)C \qquad (2)$$

BERÄKNA VILKOREN $v_C(0) = 0$ GER MED (1) & (2)

$$V_{TH} - (R_{TH}+R_3)i(0) = \underbrace{v_C(0)}_{=0}$$

$$\Rightarrow V_{TH} = (R_{TH}+R_3)A \Rightarrow A = \frac{V_{TH}}{R_{TH}+R_3} = \underline{1.6\cdot 10^{-4}}$$

SOM GER $v_C(t)$ SOM (MED (1))

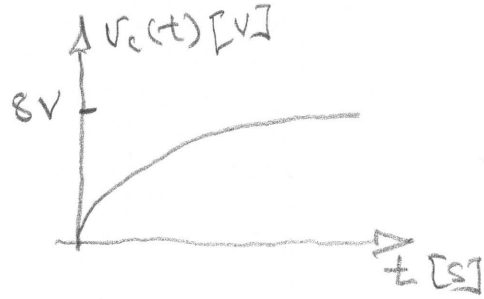
$$v_C(t) = V_{TH} - (R_{TH}+R_3)\frac{V_{TH}}{(R_{TH}+R_3)}e^{-t/\tau} =$$

$$= \underline{V_{TH}(1-e^{-t/\tau})}$$

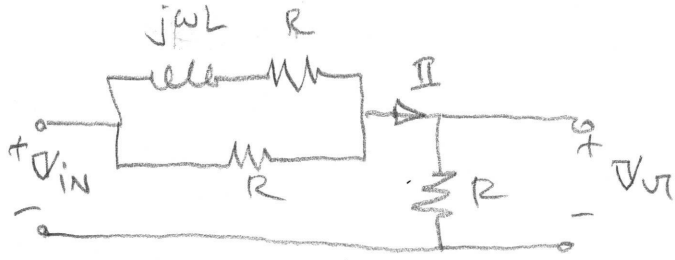
MED SIFFROR

$$v_C(t) = 8(1-e^{-2t}) \text{ VOLT}$$

SVAR: $v_C(t) = 8(1-e^{-2t})$ V



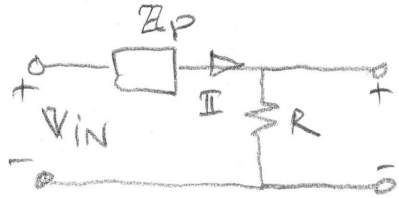
4.



P-kopplingen:

$$Z_p^{-1} = \frac{1}{j\omega L + R} + \frac{1}{R} = \frac{2R + j\omega L}{R(j\omega L + R)}$$

$$Z_p = \frac{R(j\omega L + R)}{2R + j\omega L}$$



KSPL ger

$$\begin{cases} V_{IN} - Z_p II - R II = 0 \\ V_{UT} - R II = 0 \end{cases}$$

$$\Rightarrow \frac{V_{UT}}{V_{IN}} = \frac{R}{Z_p + R} = \frac{R}{\frac{R(j\omega L + R)}{2R + j\omega L} + R} =$$

$$= \frac{R(2R + j\omega L)}{R(j\omega L + R) + R(2R + j\omega L)} =$$

$$= \frac{2R + j\omega L}{3R + 2j\omega L} = H(\omega) \qquad (1)$$

Tag beloppet av $H(\omega)$

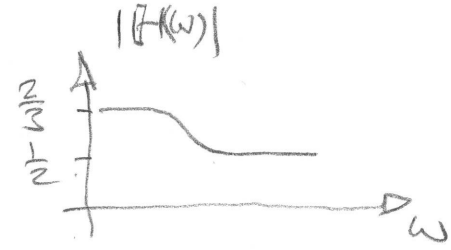
$$|H(\omega)| = \frac{\sqrt{(2R)^2 + (\omega L)^2}}{\sqrt{(3R)^2 + (2\omega L)^2}}$$

För $\omega \to \infty$

$$\lim_{\omega \to \infty} |H(\omega)| = \frac{\omega L}{2\omega L} = \frac{1}{2}$$

FÖR $\omega \to 0$

$$|H(\omega)| \underset{\omega \to 0}{=} \frac{2R}{3R} = \frac{2}{3}$$



FASFUNKTIONEN FÅS UR (1)

$$\varphi(\omega) = \arctan\left(\frac{\mathrm{Im}(H(\omega))}{\mathrm{Re}(H(\omega))}\right) \qquad (2)$$

(1) ger $H(\omega) = \dfrac{2R+j\omega L}{3R+2j\omega L} = \dfrac{(2R+j\omega L)(3R-j\omega L)}{(3R)^2+(2\omega L)^2} =$

$$= \frac{6R^2 - 4jR\omega L + 3j\omega R + 2(\omega L)^2}{(3R)^2+(2\omega L)^2} =$$

$$= \frac{6R^2+2(\omega L)^2 - jR\omega L}{(3R)^2+(2\omega L)^2}$$

i (2) ger

$$\varphi(\omega) = \arctan\left(\frac{-R\omega L}{6R^2+2(\omega L)^2}\right) \overset{\text{för } \omega L = R}{=}$$

$$= \arctan\left(\frac{-R^2}{6R^2+2R^2}\right) = \arctan\left(-\frac{1}{8}\right)$$

$$\simeq -7.125^\circ$$

SVAR

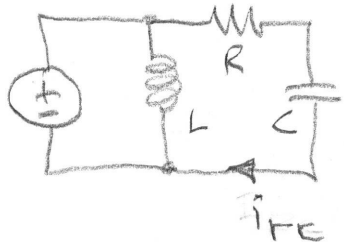
a) $H(\omega) = \dfrac{2R+j\omega L}{3R+2j\omega L}$

b) $|H(\omega \to 0)| = \frac{2}{3}$

$|H(\omega \to \infty)| = \frac{1}{2}$

c) $\varphi(\omega L = R) = -7.125^\circ$

5.



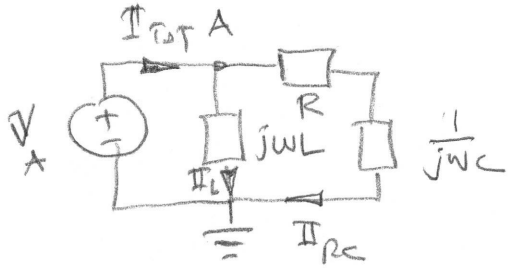
$I_{RC} = 10\angle 0°$ mA

$f = 7$ kHz

$L = 32$ mH

$R = 1.5$ kΩ

$C = 20$ nF



NODANALYS

$$V_A = \left(R + \frac{1}{j\omega C}\right) I_{RC}$$

$$= \left(1500 - \frac{j}{43982.3}\right) 10 \cdot 10^{-3} \angle 0°$$

$$= 18.82 \angle -37.16° \text{ VOLT}$$

STRÖMMEN $I_L = \frac{V_A}{Z_L} = \frac{18.82\angle -37.16°}{j\omega L} =$

$$= \frac{18.82 \angle -37.16°}{1407.43 \angle 90°} = 0.01337 \angle -127.16° \text{ A}$$

TOTALA STRÖMMEN FRÅN SP. KÄLLAN

$$I_{TOT} = I_L + I_{RC} = 13.37 \cdot 10^{-3} \angle -127.16° +$$

$$+ 10 \cdot 10^{-3} \angle 0° = 1.924 \cdot 10^{-3} - j10.65 \cdot 10^{-3} =$$

$$= 10.82 \cdot 10^{-3} \angle -79.75° \text{ A}$$

SVAR: STRÖMMEN FRÅN SPÄNNINGS KÄLLAN BLIR

$I = I_{TOT} = 10.82 \angle -79.8°$ mA

SPÄNNINGEN HOS SP. KÄLLAN

$V = V_A = 18.82 \angle -37.2°$ VOLT